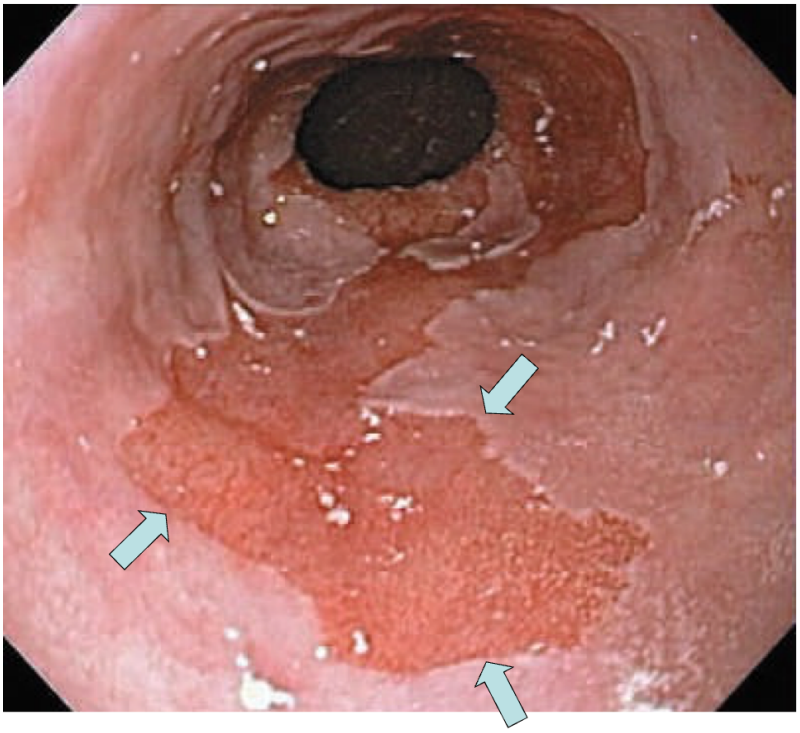
Esophageal Cancer Cases

Case 1

62M with longstanding reflux has first EGD.

Reflux got worse during training for a triathalon



Case 1

Pathology shows Barrett's metaplasia without dysplasia

What is appropriate follow up?

Biopsy strategy?

Case 1 Barrett's metaplasia without dysplasia

What is appropriate follow up?

AGA Guidelines

- No dysplasia: 3-5 years
- Low grade dysplasia: 6-12 months
- High grade dysplasia 3 months
  - (in the absence of ablation)

Case 1 Barrett's metaplasia without dysplasia

Biopsy strategy?

AGA Guidelines

- White light endoscopy
- 4-quadrant biopsy every 2cm
- Mucosal irregularity biopsied separately
- 4-quadrant biopsy every 1cm if dysplasia)

Case 2

Pathology shows high-grade dysplasia

Treatment Options:

- Observation
- Esophagectomy
- Cryotherapy
- Irreversible Electroporation
- Radio-frequency Ablation

Observation

You receive a hand-written note from the family
inviting you to the funeral of the patient
who passed after a heroic battle with esophageal cancer

Case 2

Esophagectomy

Correct answer, wrong century (not the 20th)

Case 2

Cryotherapy

Correct answer, wrong century (not the 22nd)

Case 2

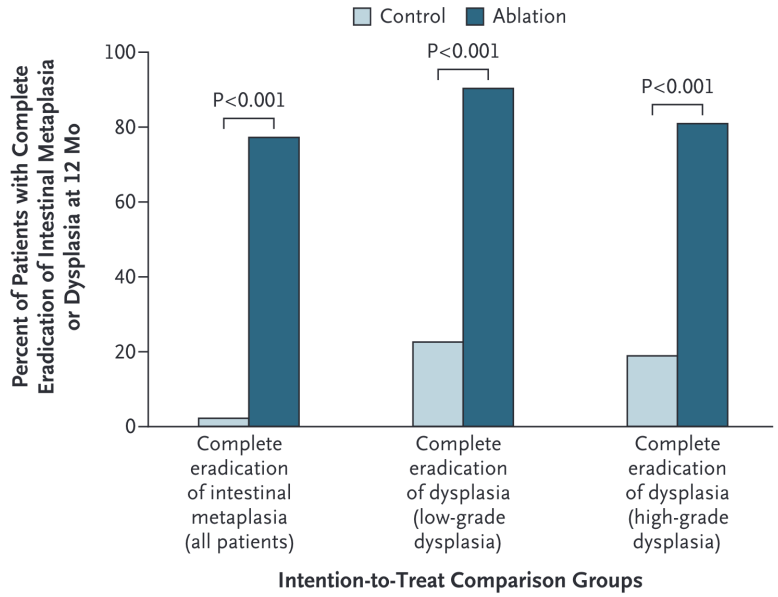
Irreversible Electroporation



Case 2

Radio-frequency Ablation

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year



Case 3

(Shaheen et al. 2009)

Case 3

EGD shows a 21mm nodule in the Barrett's esophagus

Treatment Options:

- Surveillance
- Minimally Invasive Esophagectomy
- **Endoscopic Mucosal Resection**
- [Endoscopic Submucosal Dissection]
- Radio-frequency Ablation Barxx

See Forbes - ASGE guidelines - ESD for lesions >2cm

Surveillance

You receive a hand-written note from the family
inviting you the patient's 70th birthday
after being treated at a competing medical center
for esophageal cancer

Case 3

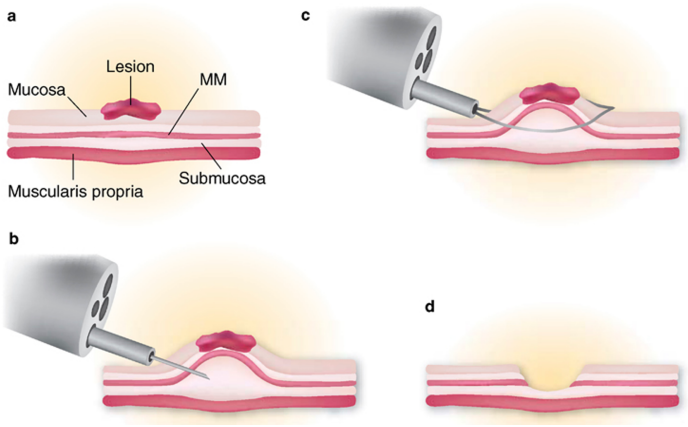
Minimally Invasive Esophagectomy

Correct answer, wrong patient

Case 3

Endoscopic Mucosal Resection

Endoscopic procedure resects mucosal tumor



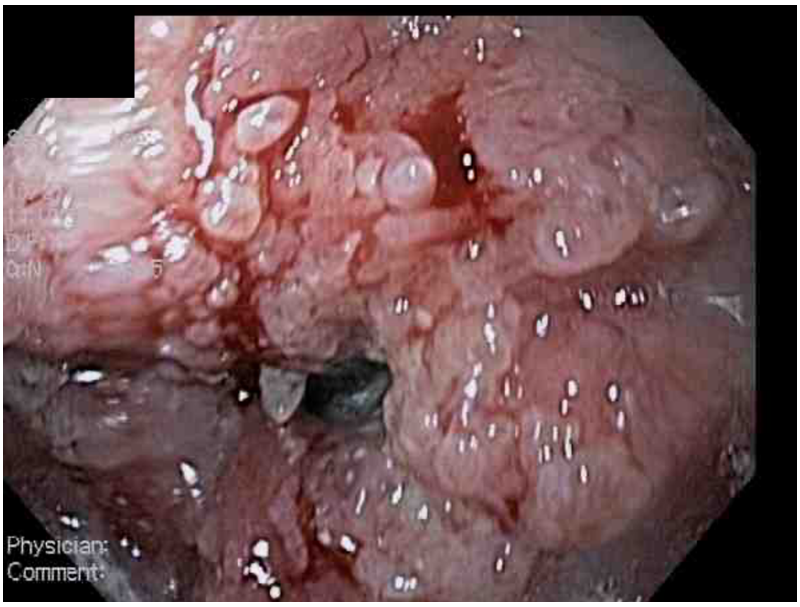
Case 4

Radio-frequency Ablation Barxx

Correct answer, wrong question

Case 4

Your patient from Case 2 returns with dysphagia and weight loss



Case 4 Adenocarcinoma

Pathology shows adenocarcinoma

Workup?

Case 4 Adenocarcinoma

Workup?

- Barium Swallow
- Endoscopic Ultrasound
- CT Chest/Abdomen/Pelvis

Barium Swallow

Correct answer, wrong century

Case 4

Endoscopic Ultrasound

You receive a hand-written note from the family

inviting you the patient's funeral

after they died from an esophageal perforation

which occurred during EUS

Autopsy showed T3 adenocarcinoma

[EUS in Patients with Dysphagia]

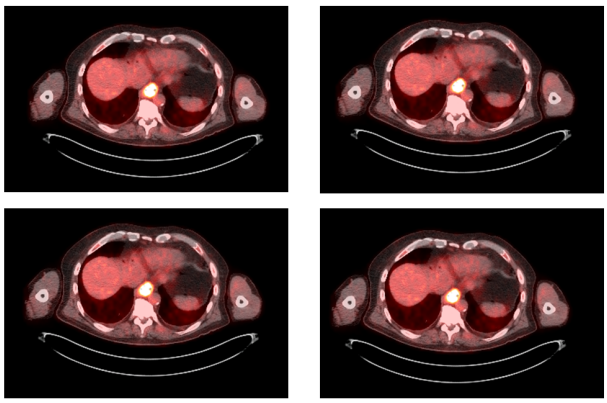
Case 4

CT Chest/Abdomen/Pelvis

What test do you order next?

PET scan

A PET scan is most accurate method of staging esophageal cancer



T3 N0 M0 adenocarcinoma

Treatment Options - First Treatment Course

- MI Esophagectomy
- Chemo + Radiation
- Chemotherapy

Patient

60M with GERD abd Barrett's presents with food impaction. EGD shows a 2cm ulcerated nodular lesion. EUS shows invasion of muscularis propria

- ESD
- Ablation
- Cryo
- PET and surgical consultation
-

MI Esophagectomy

Correct answer, wrong timing

Why?

T3 N0 M0 adenocarcinoma

Chemo + Radiation

Concurrent chemotherapy and radiation followed by surgery = Trimodality therapy

CROSS Trial

- 368 esophageal cancer patients randomized:
  - Surgery alone
  - Chemo+RT → Surgery
- 75% adenocarcinoma
- T3: 80%. T2: 17%
- age $\tilde{x}$=60
- longer survival with Chemo+RT → Surgery

CROSS Trial Details

Chemotherapy: Weekly carboplatin and paclitaxel Radiation: 4140 cGy in 23 fractions (180cGy/fraction)

(Shapiro et al. 2015)

CROSS - Overall Survival

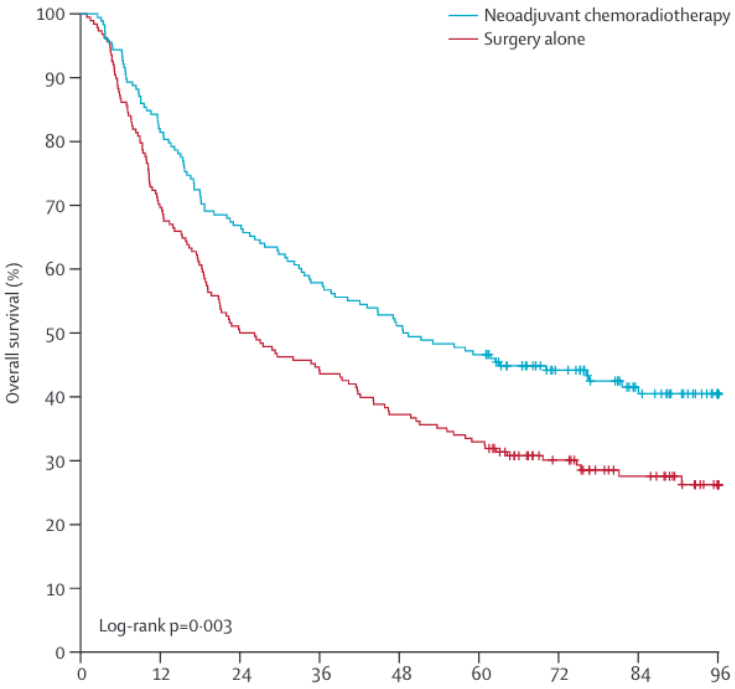


Figure 1: Surgery vs ChemoRT → Surgery

(Shapiro et al. 2015)

CROSS - Survival by Histology

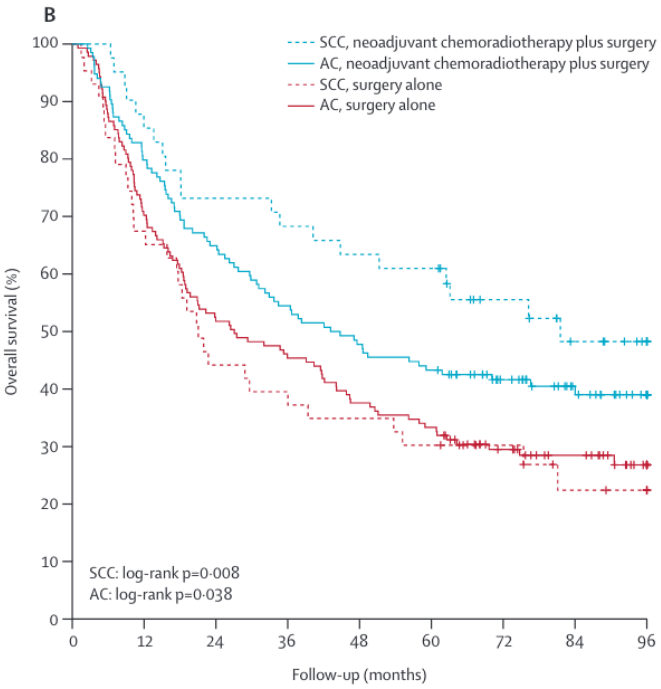


Figure 2: Surgery vs ChemoRT → Surgery

CROSS - Adenocarcinooma

(Shapiro et al. 2015)

Median survival 43mo vs 27mo

Pathologic complete response in 23%

CROSS - Squamous cell carcinoma

Median survival 82mo vs 21mo

Pathologic complete response in 40%

(Shapiro et al. 2015)

T3 N0 M0 Adenocarcinoma

Family asks if there is a better treatment option than CROSS

Chemotherapy

"Sandwich" Chemotherapy may be superior to Trimodality therapy

EsoPEC Trial

- Adenocarcinoma esophagus - T1 N+ or T2-4a M0. Median age =63. 89% men
- Randomized to CROSS (n=217) vs FLOT chemotherapy (n=221) = 438
- CROSS: carboplatin/paclitaxel + 4140cGy → Surgery
- FLOT: FLOT → Surgery → FLOT
- Excluded: Squamous cell, gastric cancer, T1N0, T4b, M1

EsoPEC Trial Results

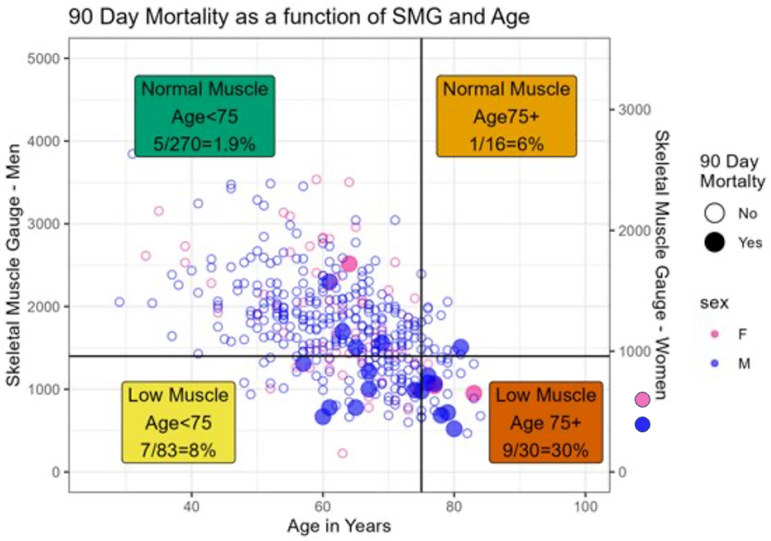
- Surgery performed in 371/438 patients
- 90-day mortality 4.3% (3.2% in FLOT and 5.6% CROSS)
- Median survival 66mo in FLOT arm and 37mo in Cross arm
- 3-year overall survival 57% FLOT vs 51% CROSS
- 5-year overall survival 51% FLOT vs 29% CROSS
- pCR 17% for FLOT and 10% CROSS

Case 7

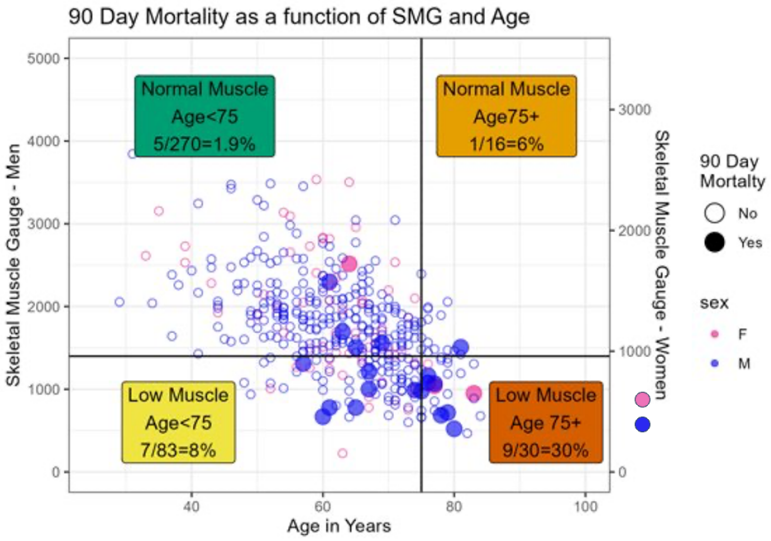
Patient returns after FLOT chemotherapy

What are risks of surgery?

Preoperative Evaluation



Preoperative Evaluation



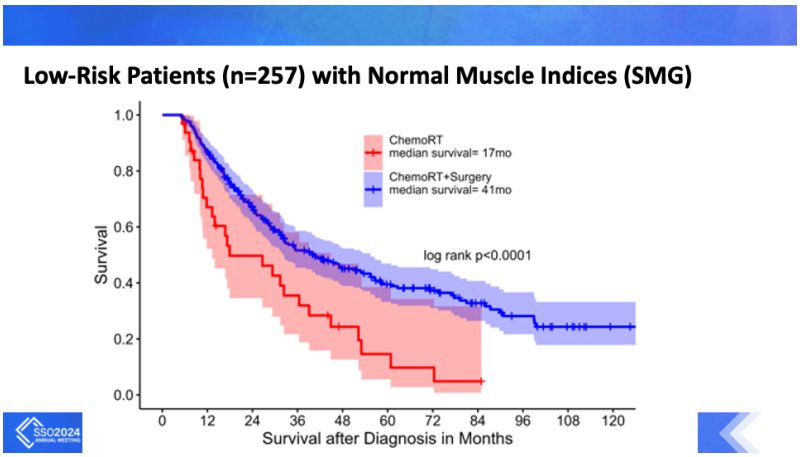
25th percentile: Grip Strength 26kg (men) / 16kg (women)

Case 7

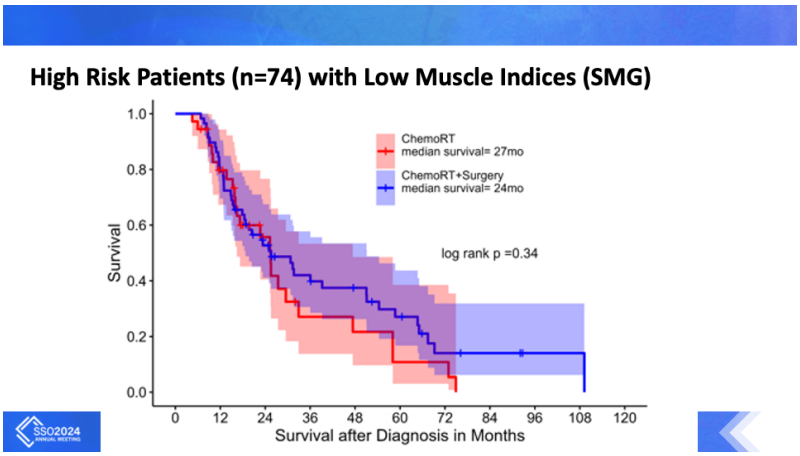
Patient returns after FLOT chemotherapy

What are surgical options?

Low Risk Adenocarcinoma



High Risk Adenocarcinoma



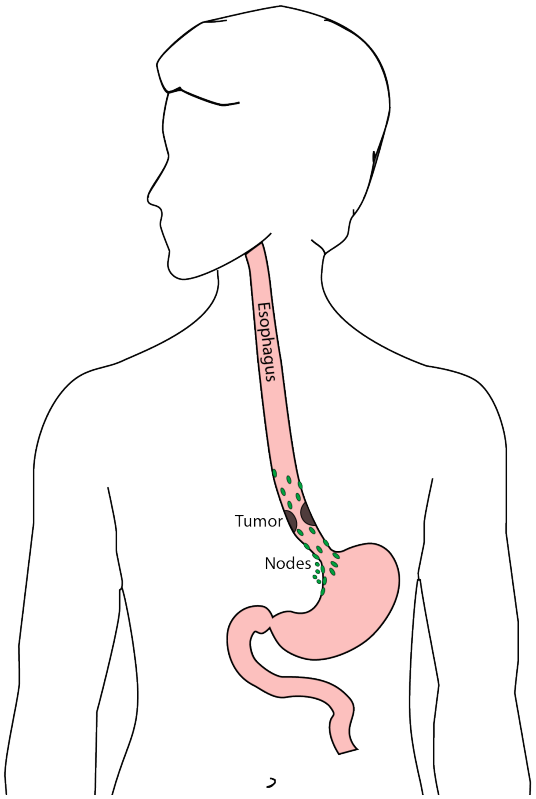
Surgery for Esophageal Cancer

Surgery for esophageal cancer is performed for:

- Superficial Tumors (T1) not completely removed by endoscopy
- Localized Tumors (T2N0)
- Locally Advanced (T3) after preoperative therapy.

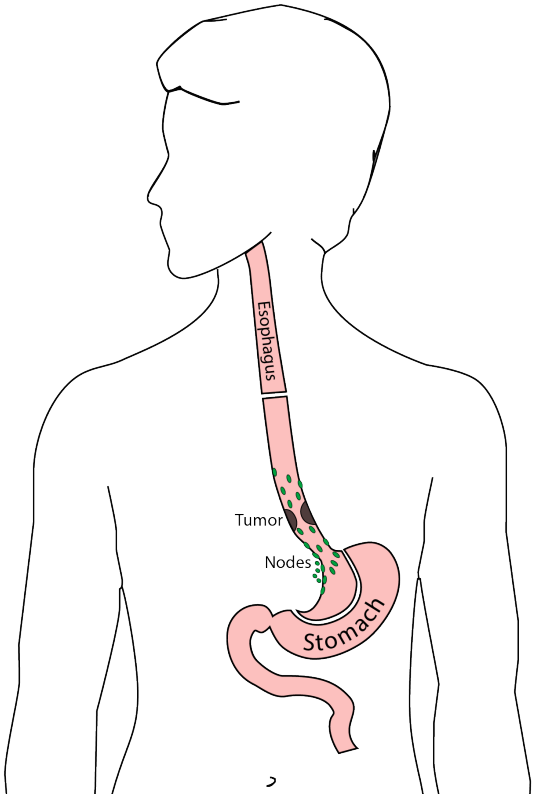
Goals of Surgery

- Remove tumor from esophagus
- Remove surrounding lymph nodes
- Create a new esophagus



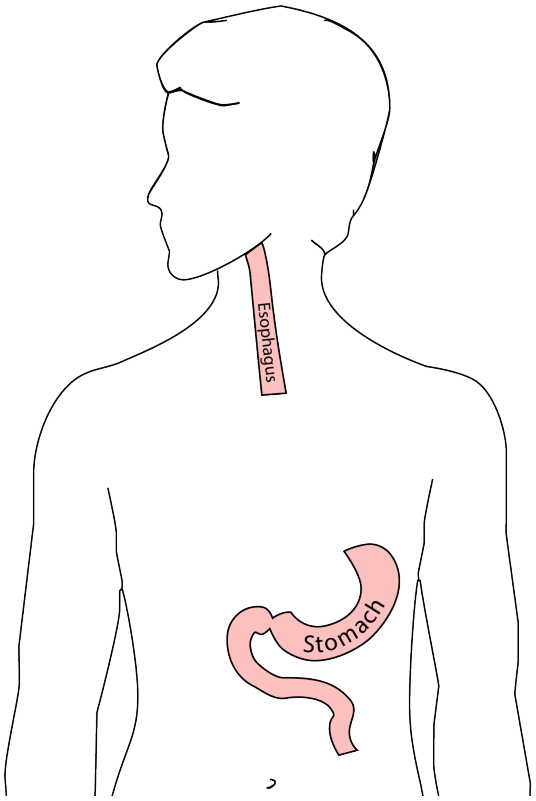
Ivor Lewis (Transthoracic) Esophagectomy

- Removes tumor
- Removes lower 1/3 of esophagus
- Removes surrounding lymph nodes
- Reconstruction of GI tract



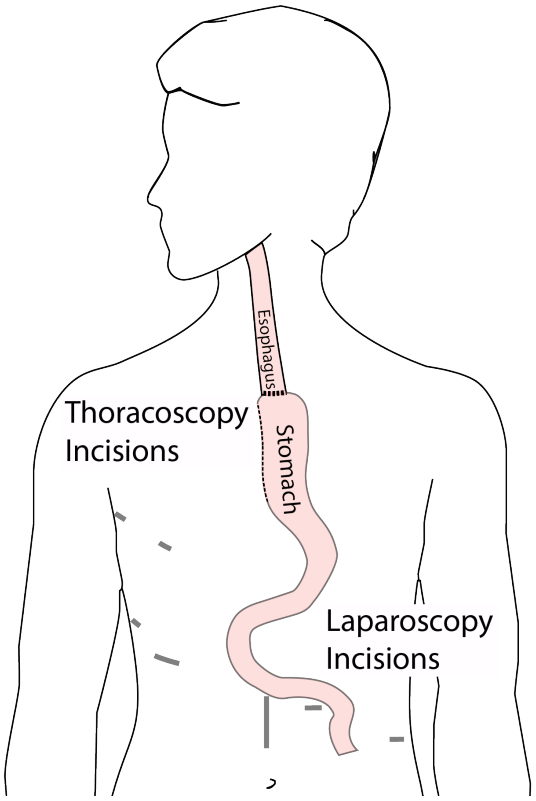
Reconstruction

A new esophagus is created from the stomach in the abdomen by fashioning it into a tube.

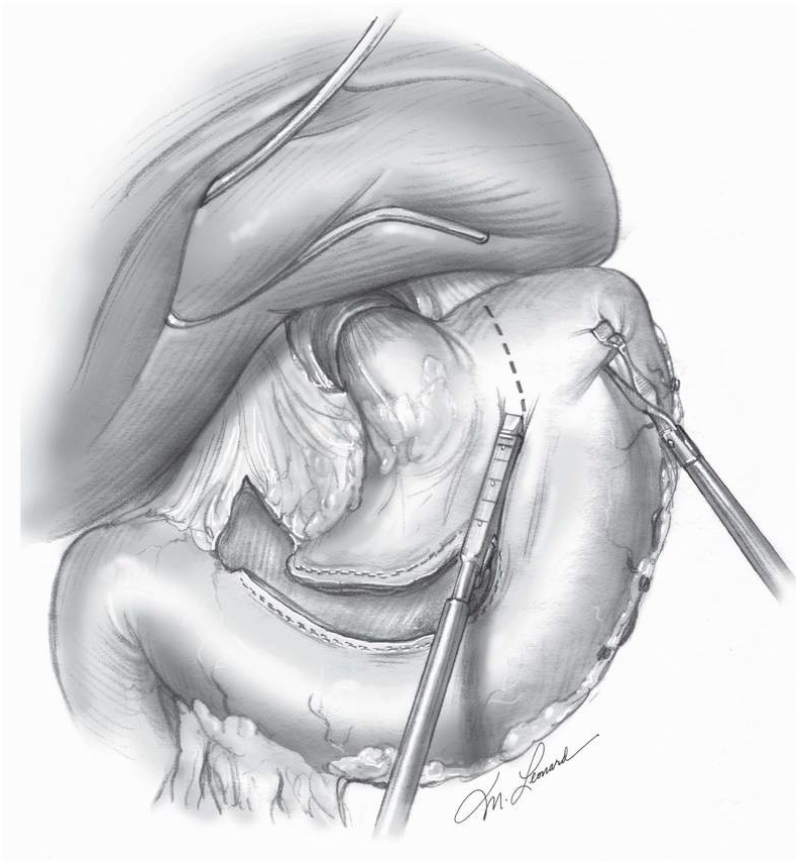


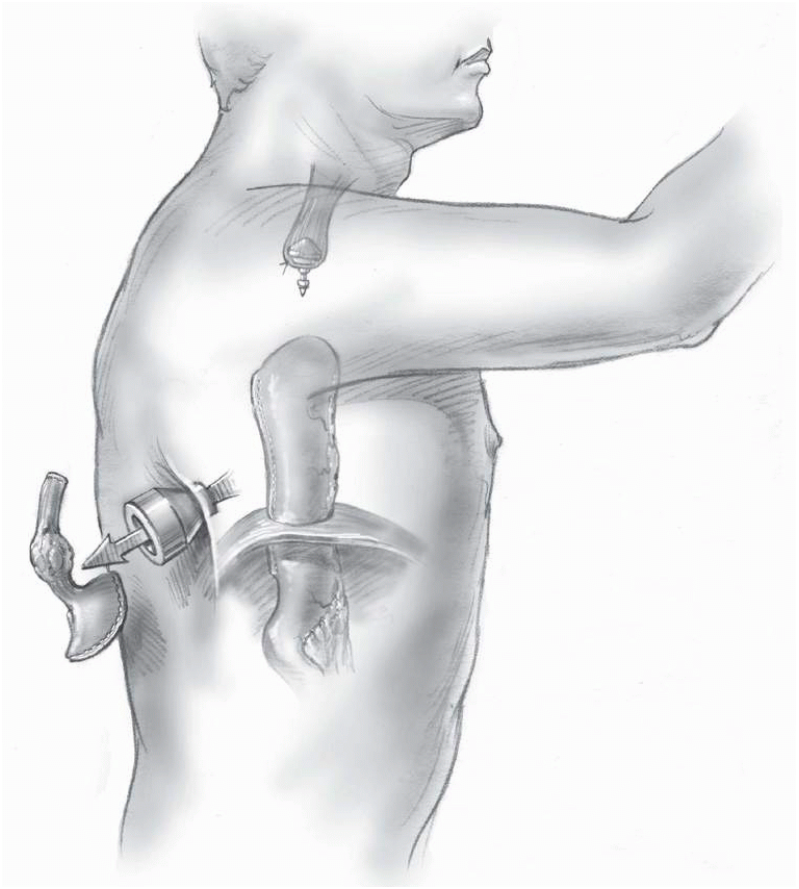
Minimally-invasive Ivor Lewis

- Laparoscopic mobilization of stomach
- Construction of gastric conduit
- Thoracic anastomosis

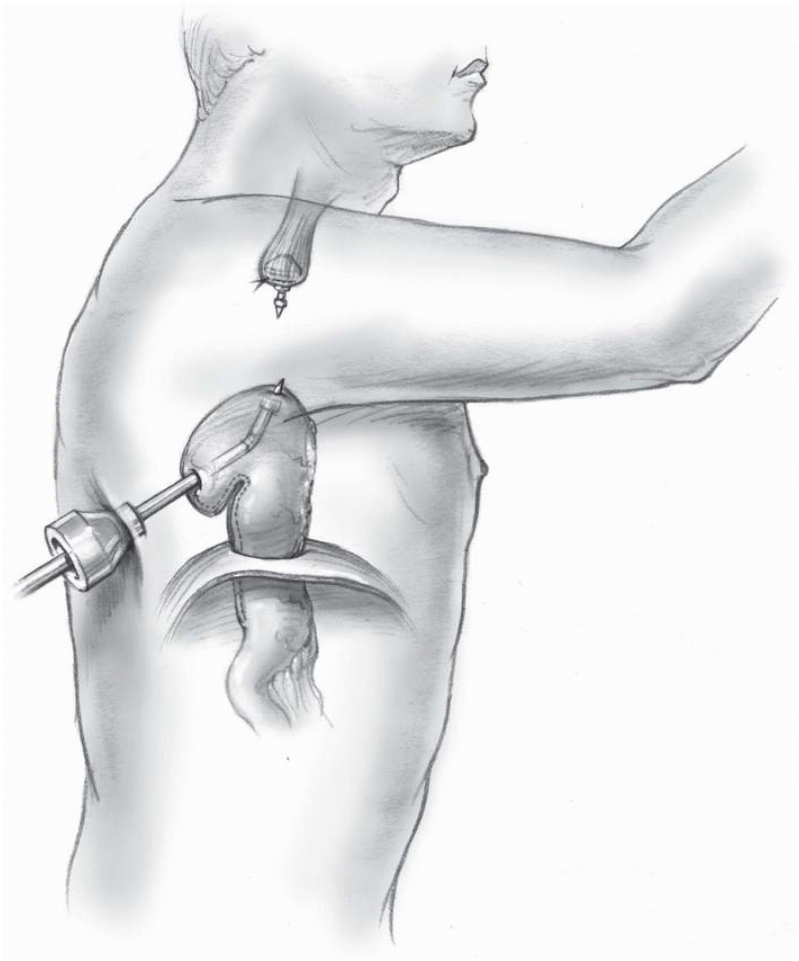

Esophagus
Thoracoscopy
Incisions
Stomach
Laparoscopy
Incisions

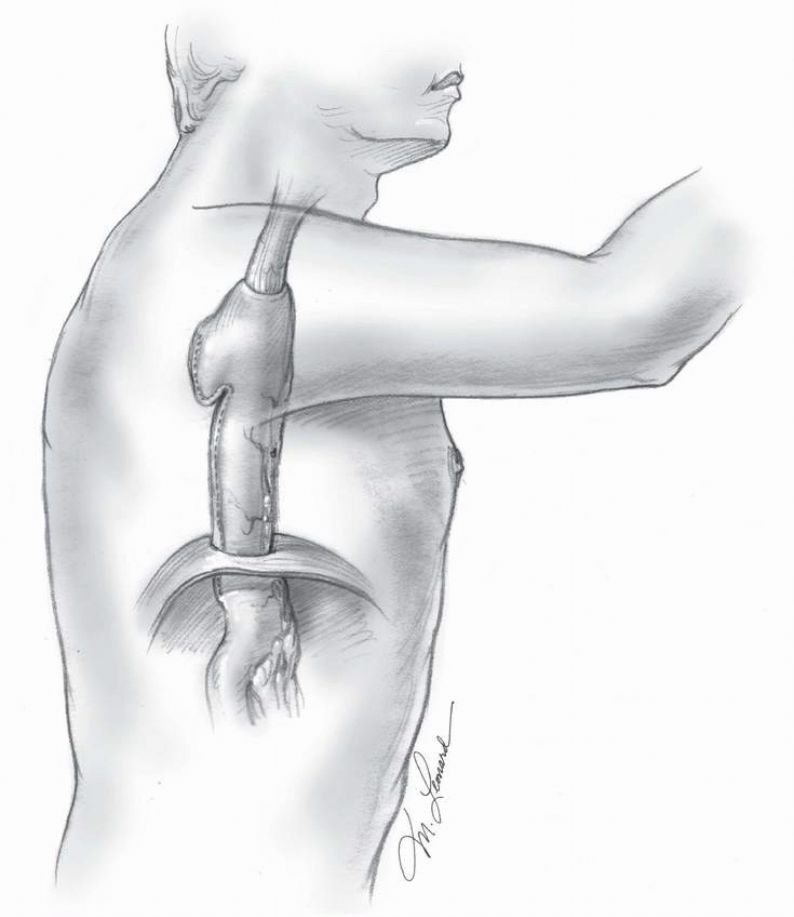
Ivor Lewis esophagectomy





Ivor Lewis esophagectomy



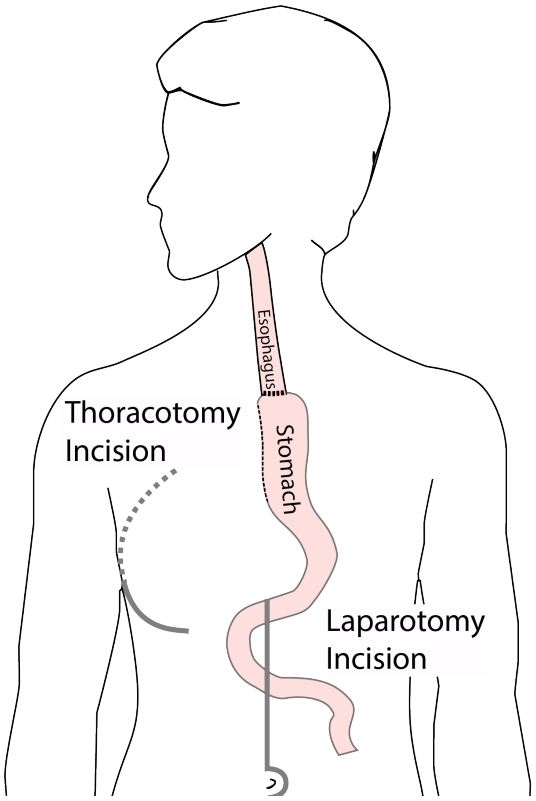


Thoracic Circular Stapled Anastomosis

Open Ivor Lewis

We use the mininally-invasive approach in 95% of cases

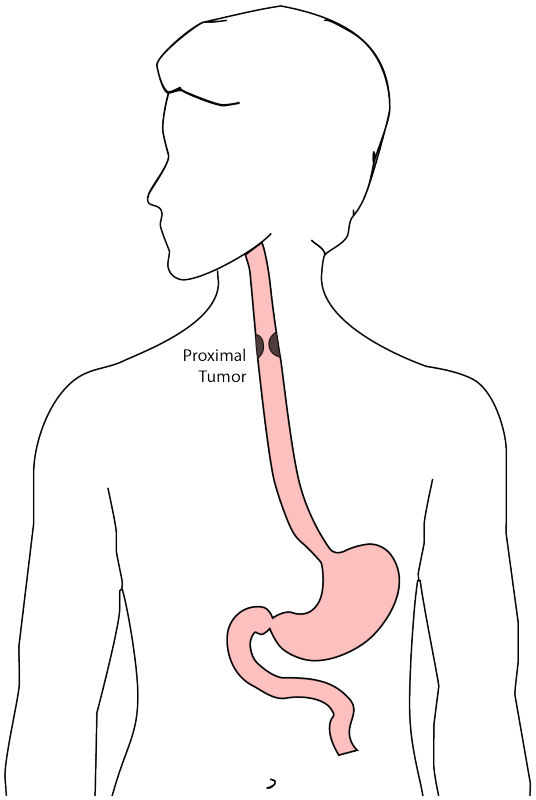
In some cases, an open approach is still necessary.



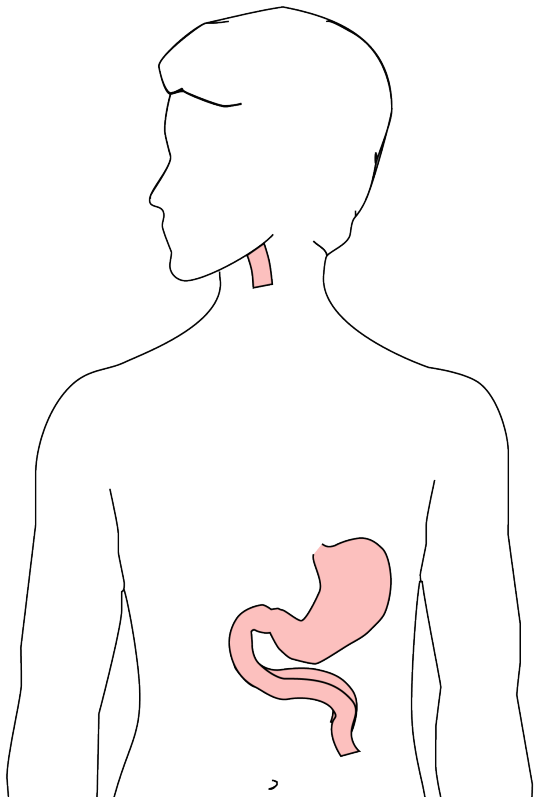
Total Esophagectomy

For patients with tumors in the upper esophagus, we need to remove more of the esophagus

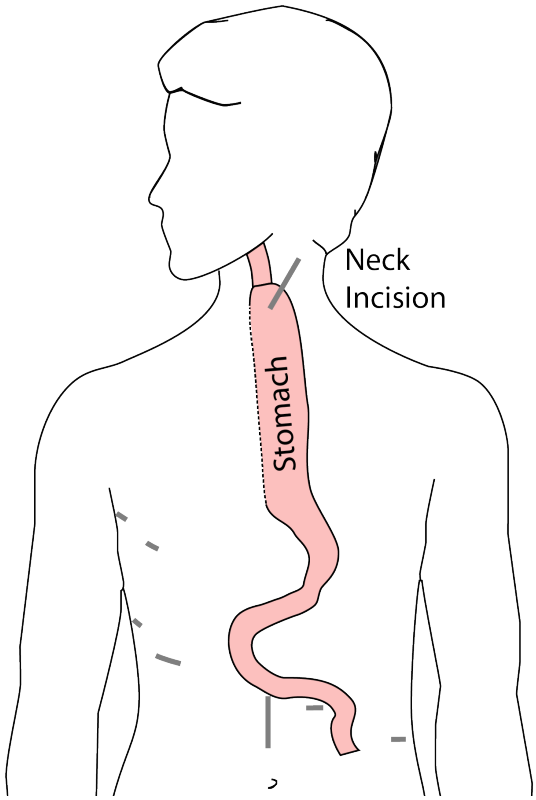
We need to remove the whole esophagus, including the portion in the neck


Proximal
Tumor

McKeown Esophagectomy

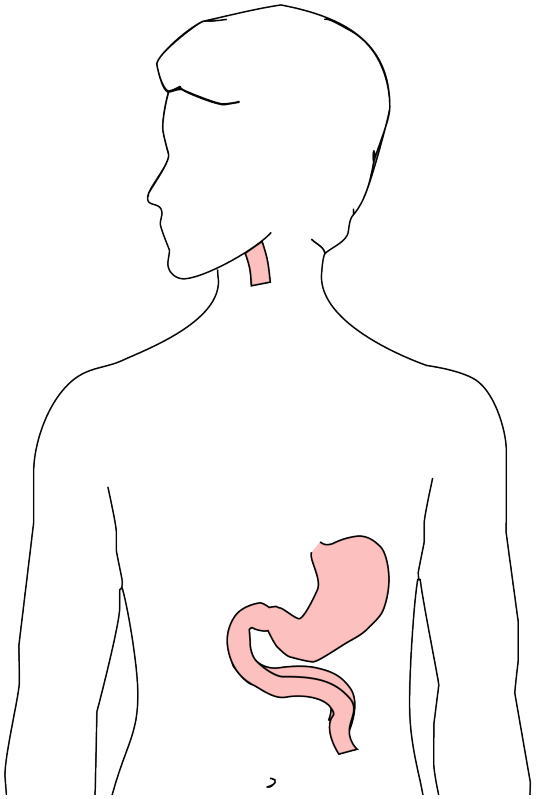


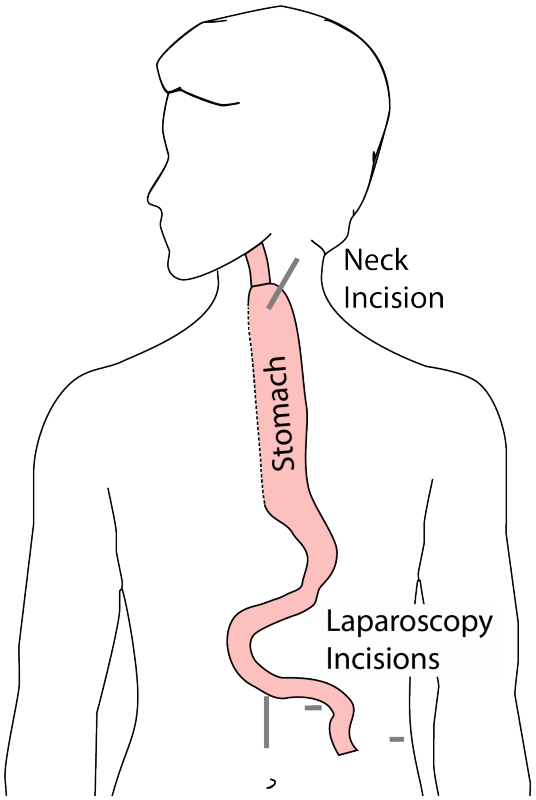
All of esophagus removed



Connection made in the neck

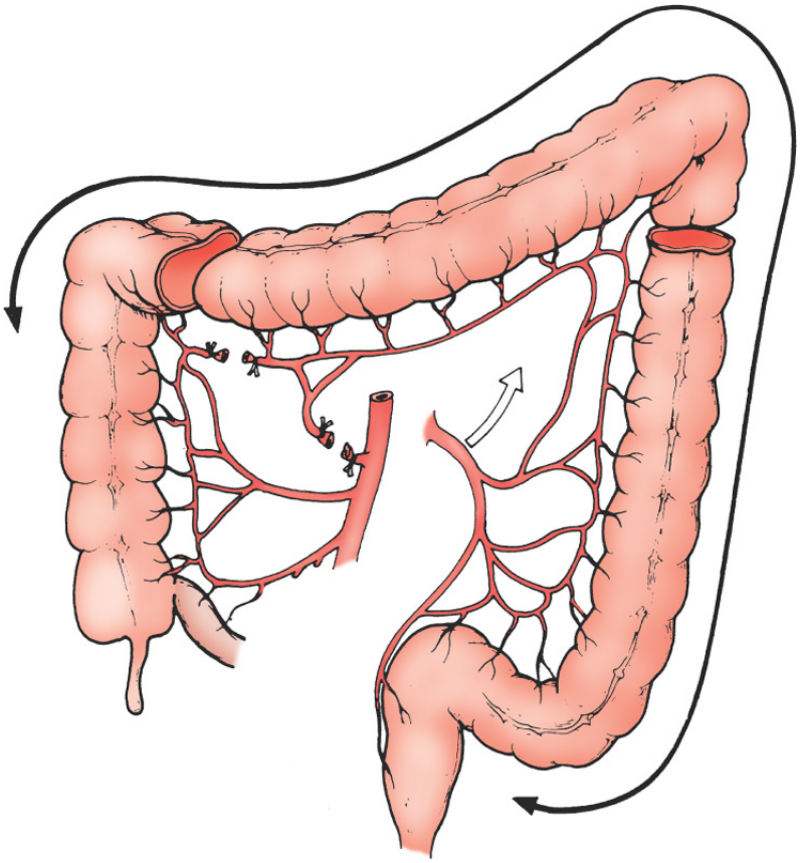
Transhiatal Esophagectomy





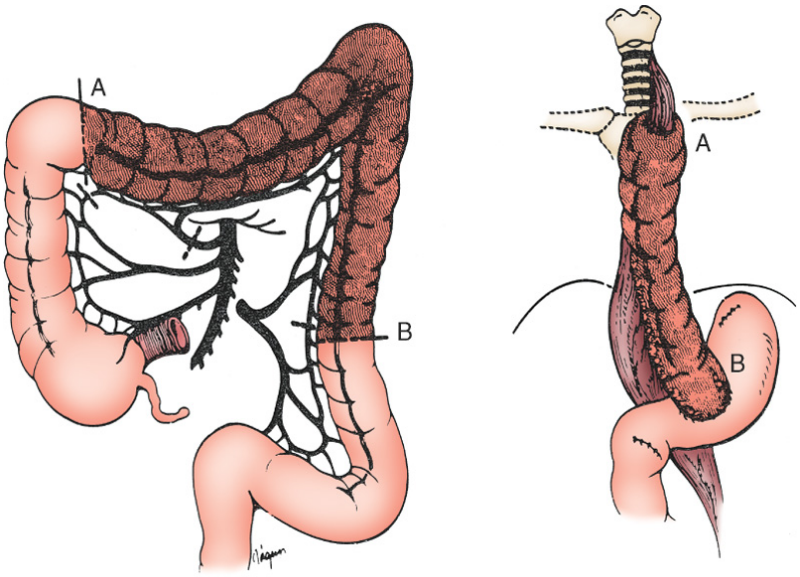
Colon Interposition

If the stomach is not suitable to make a new esophagus, the colon can be used to replace the esophagus



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Colon Interposition



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Postop nutrition

Postop patient has macrocytic anemia with Hb 8.5 and MCV 110fL. Which is most likely nutritonal deficit?

- Vitamin B1
- Vitamin B6
- Vitamin B12
- Vitamin C
- Vitamin D

Orientation Handbook



References

Shaheen, Nicholas J., Prateek Sharma, Bergein F. Overholt, et al. 2009. "Radiofrequency Ablation in Barrett's Esophagus with Dysplasia." *The New England Journal of Medicine* 360 (22): 2277–88. https://doi.org/10.1056/NEJMoa0808145.

Shapiro, Joel, J. Jan B. van Lanschot, Maarten C. C. M. Hulshof, et al. 2015. "Neoadjuvant Chemoradiotherapy Plus Surgery Versus Surgery Alone for Oesophageal or Junctional Cancer (CROSS): Long-Term Results of a Randomised Controlled Trial." *The Lancet. Oncology* 16 (9): 1090–98. https://doi.org/10.1016/S1470-2045(15)00040-6.